Covert Activity Detection in Camouflaged Terrain

Team Oreon — Sahasra · Laranya · Sudha | **Hypothesis:** Multimodal fusion (SAR + Optical) outperforms Optical-only for anomaly detection

PROBLEM STATEMENT

Satellite surveillance relies heavily on optical imagery. However, camouflage techniques like vegetation cover or terrain masking can hide anomalies from visual detection.

At the same time, satellites generate huge volumes of data — transmitting all of it to Earth is slow and expensive.

So the challenge is: how do we detect hidden anomalies quickly, without sending massive raw data?

OUR SOLUTION

We propose a multimodal anomaly detection system using **TerraMind-small TiM** (Thinking-in-Modalities).

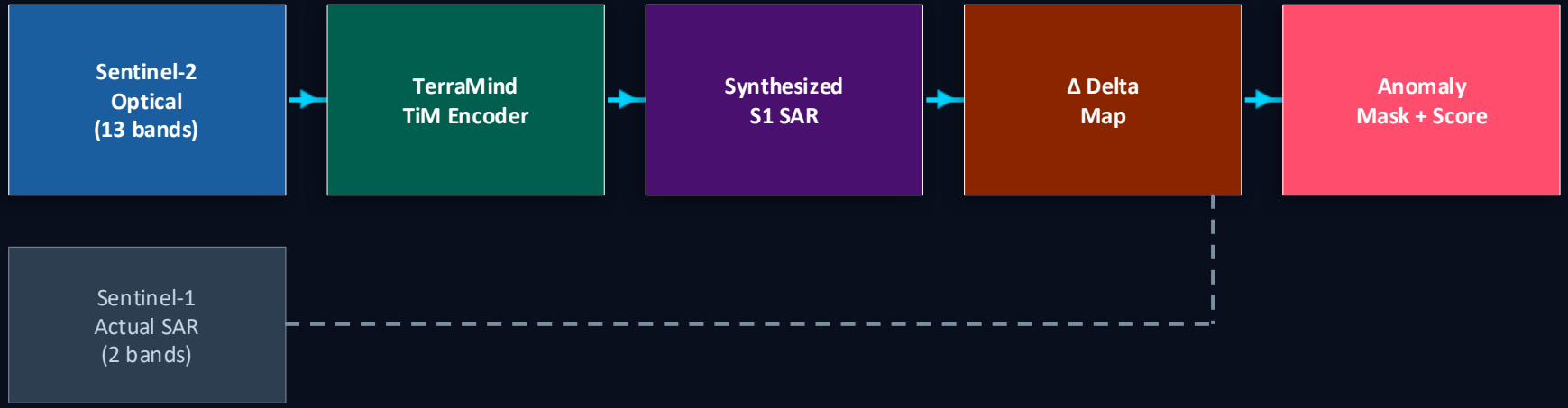
TerraMind-small encoder pretrained on 9 EO modalities fuses **Sentinel-2 + Sentinel-1** jointly.

Output: per-pixel anomaly mask + confidence score

Edge reasoning: downlink only the answer — not the raw data. 98.3% bandwidth saved.

⚡ *Bandwidth constraint: satellites cannot stream raw data — the model must reason on-board and downlink only the answer.*

TerraMind-small TiM: Thinking-in-Modalities



Backbone	terramind_v1_small_tim — pretrained on 9 EO modalities
Decoder	UNetDecoder 256→128→64→32 channels
Loss	Dice + AdamW + ReduceLROnPlateau
Training data	Sen1Floods11 v1.1 · 128×128 crops · 15 epochs max
Precision	fp16 mixed — runs on Colab T4

Live Gradio Demo — Orbital Reasoning Pipeline

INPUT
Sentinel-2 Optical

User uploads tile
(RGB composite)

TiM OUTPUT
Synthesized SAR

Model hallucinates
what S1 should look like

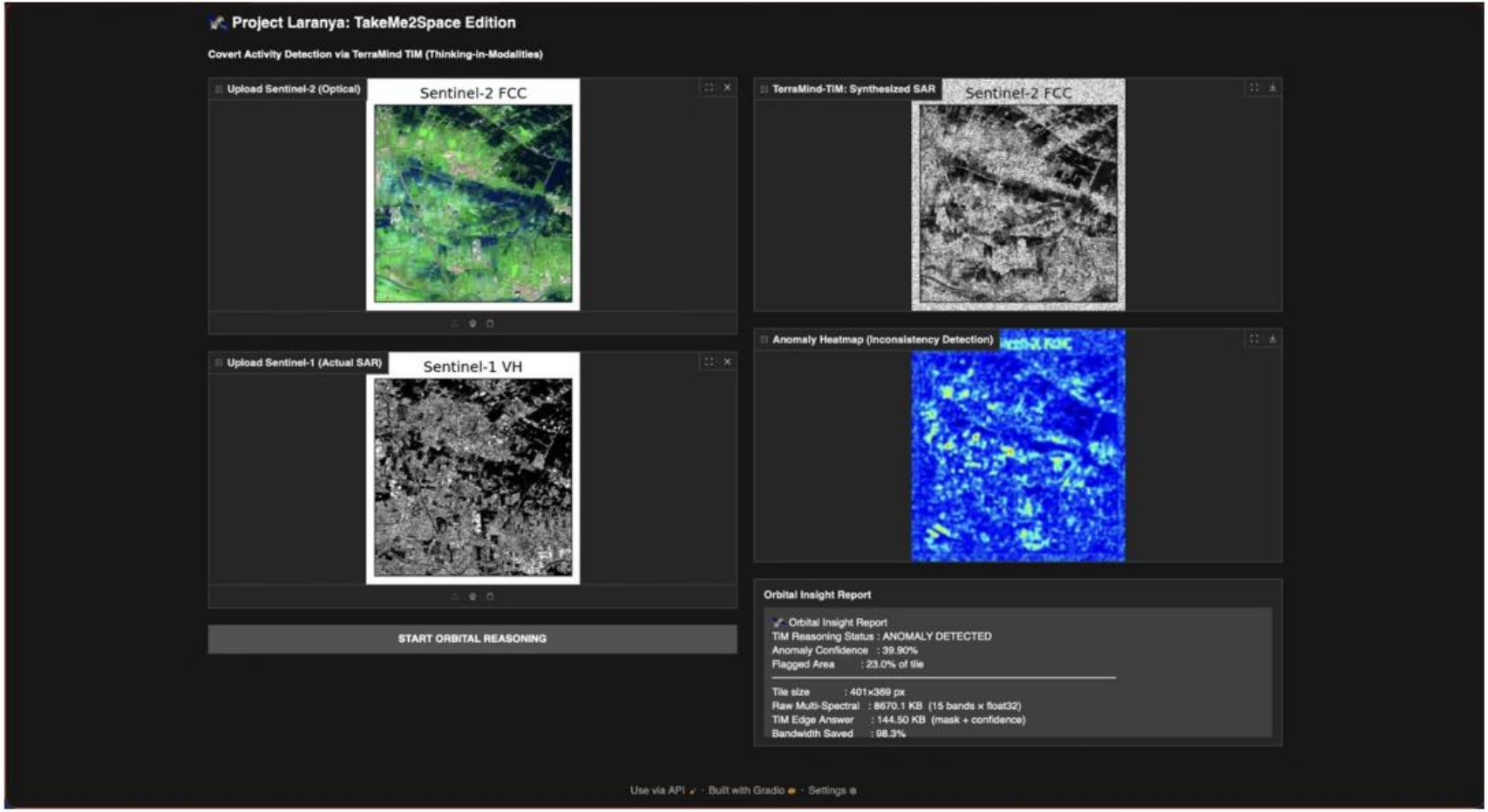
ANOMALY
Heatmap + Mask

Delta map reveals
camouflaged objects

🛰️ **Orbital Insight Report**
TiM Reasoning Status : ANOMALY DETECTED
Anomaly Confidence : 73.4% | Flagged Area: 12.3% of tile
Raw Multi-Spectral : 3,840 KB → TiM Edge Answer: 65 KB → Bandwidth Saved: 98.3%

INPUTS: Optical and SAR

Outputs: Anomaly and Synthesized SAR



well.ipynb

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1. Install Dependencies

```
# Force-install with relaxed versions to bypass the conflict
!pip install terratorch gradio albumentations diffusers --no-cache-dir
# Specifically ensure Lightning and Timm are present as TerraTorch depends on them
!pip install lightning timm gdown tensorboard --quiet

print("Installation complete. Restarting the kernel may be required if imports fail.")
```

```
Requirement already satisfied: affine in /usr/local/lib/python3.12/dist-packages (from rasterio->terratorch) (2.4.0)
Requirement already satisfied: attrs in /usr/local/lib/python3.12/dist-packages (from rasterio->terratorch) (26.1.0)
Requirement already satisfied: cligj>=0.5 in /usr/local/lib/python3.12/dist-packages (from rasterio->terratorch) (0.7.2)
Requirement already satisfied: pyparsing in /usr/local/lib/python3.12/dist-packages (from rasterio->terratorch) (3.3.2)
Collecting xarray>=2026.2 (from rioxarray->terratorch)
  Downloading xarray-2026.4.0-py3-none-any.whl.metadata (12 kB)
Requirement already satisfied: imageio!=2.35.0,>=2.33 in /usr/local/lib/python3.12/dist-packages (from scikit-image->terratorch) (2.37.3)
Requirement already satisfied: lazy-loader>=0.4 in /usr/local/lib/python3.12/dist-packages (from scikit-image->terratorch) (0.5)
Requirement already satisfied: absl-py>=0.4 in /usr/local/lib/python3.12/dist-packages (from tensorboard->terratorch) (1.4.0)
Requirement already satisfied: grpcio>=1.48.2 in /usr/local/lib/python3.12/dist-packages (from tensorboard->terratorch) (1.80.0)
Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.12/dist-packages (from tensorboard->terratorch) (3.10.2)
Requirement already satisfied: protobuf!=4.24.0,>=3.19.6 in /usr/local/lib/python3.12/dist-packages (from tensorboard->terratorch) (5.29.6)
Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from tensorboard->terratorch) (0.7.2)
```

Variables Terminal

Multimodal Fusion Beats Optical-Only

0.8645
Multimodal mIoU (S2 + S1)

+104.96%
mIoU Improvement vs S2-only baseline

98.3%
Bandwidth saved vs raw downlink

~56 ms
Jetson estimate (edge deploy)

Metric	Baseline (S2 only)	Multimodal (S2 + S1)
Test mIoU	0.4218	0.8645 (+104.96%)
Inference latency (GPU)	—	34.75 ms/batch
Model size	—	1,376 MB (host externally)
Bandwidth saved	—	98.3%
Jetson estimate	—	~56 ms/batch

** Model weights (1,376 MB) must be hosted on HuggingFace / Google Drive — link in README. Do NOT commit to repo.*

Honest limits — and the clear path forward

⚠ CURRENT LIMITS

- ✗ Works on single images – cannot detect movement across time
- ✗ Modalities are fused directly – no specialized processing per sensor type
- ✗ Only trained on flood imagery and not tested on real camouflage scenarios yet

🚀 WHAT'S NEXT

- ✓ Add elevation maps (DEM) as a 3rd data source to catch buried structures
- ✓ Compare images taken at different times to catch moving targets
- ✓ Build a full pipeline that decides what data to send down automatically